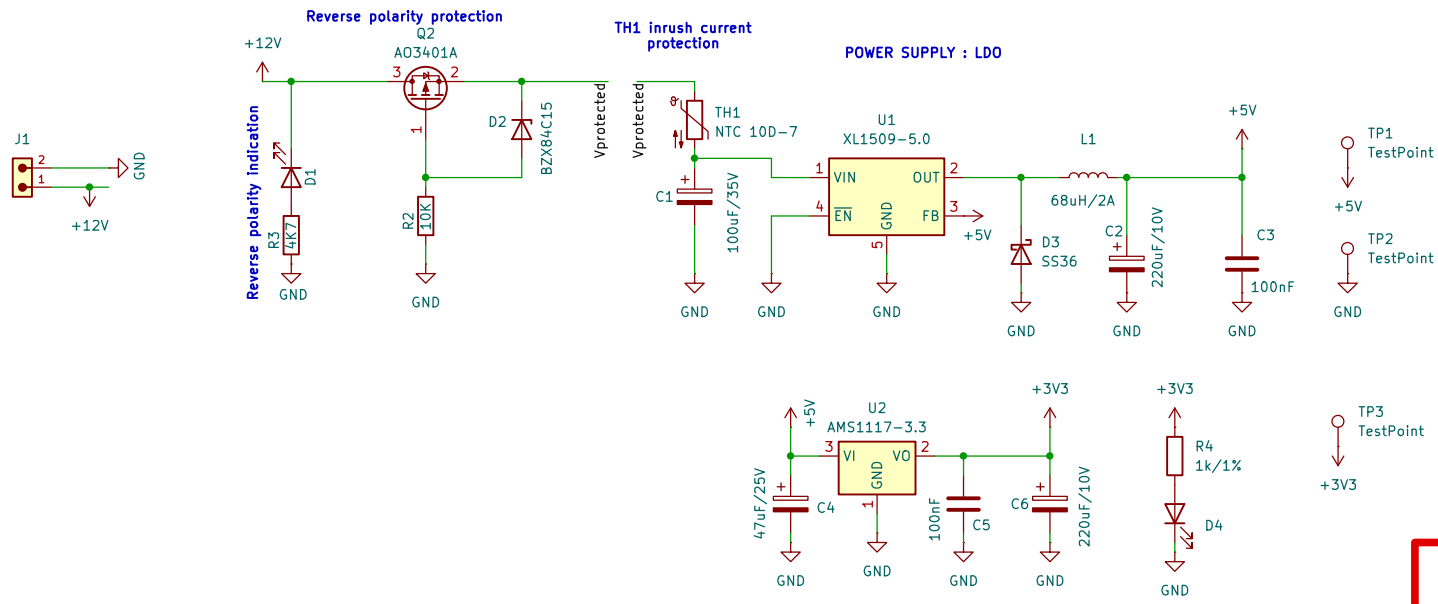






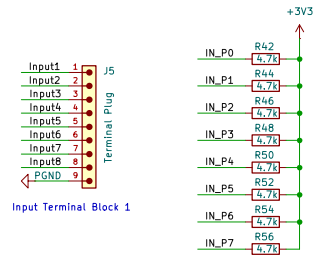
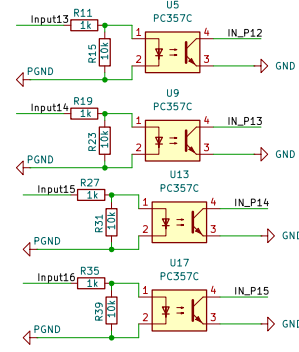
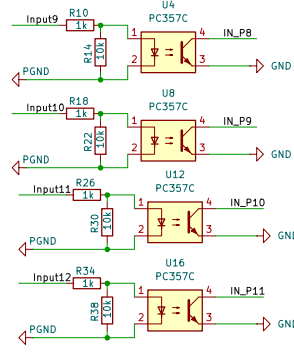
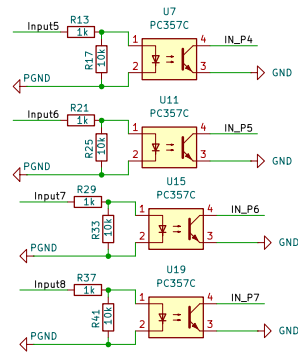
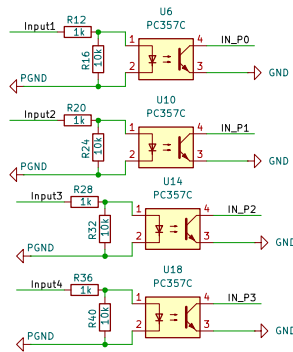
GLOBAL PINS / LABELS

- +12V
- +5V
- +3V3
- GND

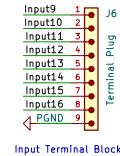
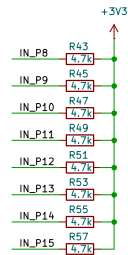
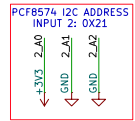
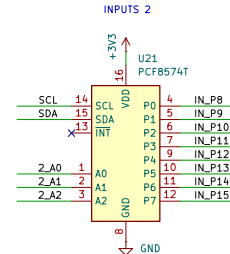
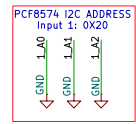
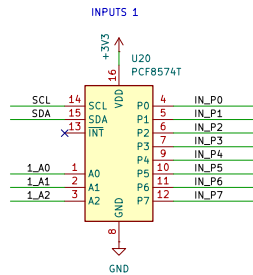
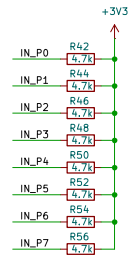
Designed by: Lucas Tivana		
DC POWER SUPPLY		
Input DC 12/24V ->5V and 3V3 DC Output		
Tivana Holdings		
Sheet: /Power Supply/		
File: Power_Supply.kicad_sch		
Title: TSC BW 16 V3.1		
Size: A4	Date: 2026-04-20	Rev: V3.1
KiCad E.D.A. 10.0.5		Id: 2/17

Id: 3/17

16 DIGITAL ISOLATED INPUTS

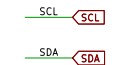


Input Terminal Block 1



Input Terminal Block 2

GLOBAL PINS / LABELS



Designed by: Lucas Tivana

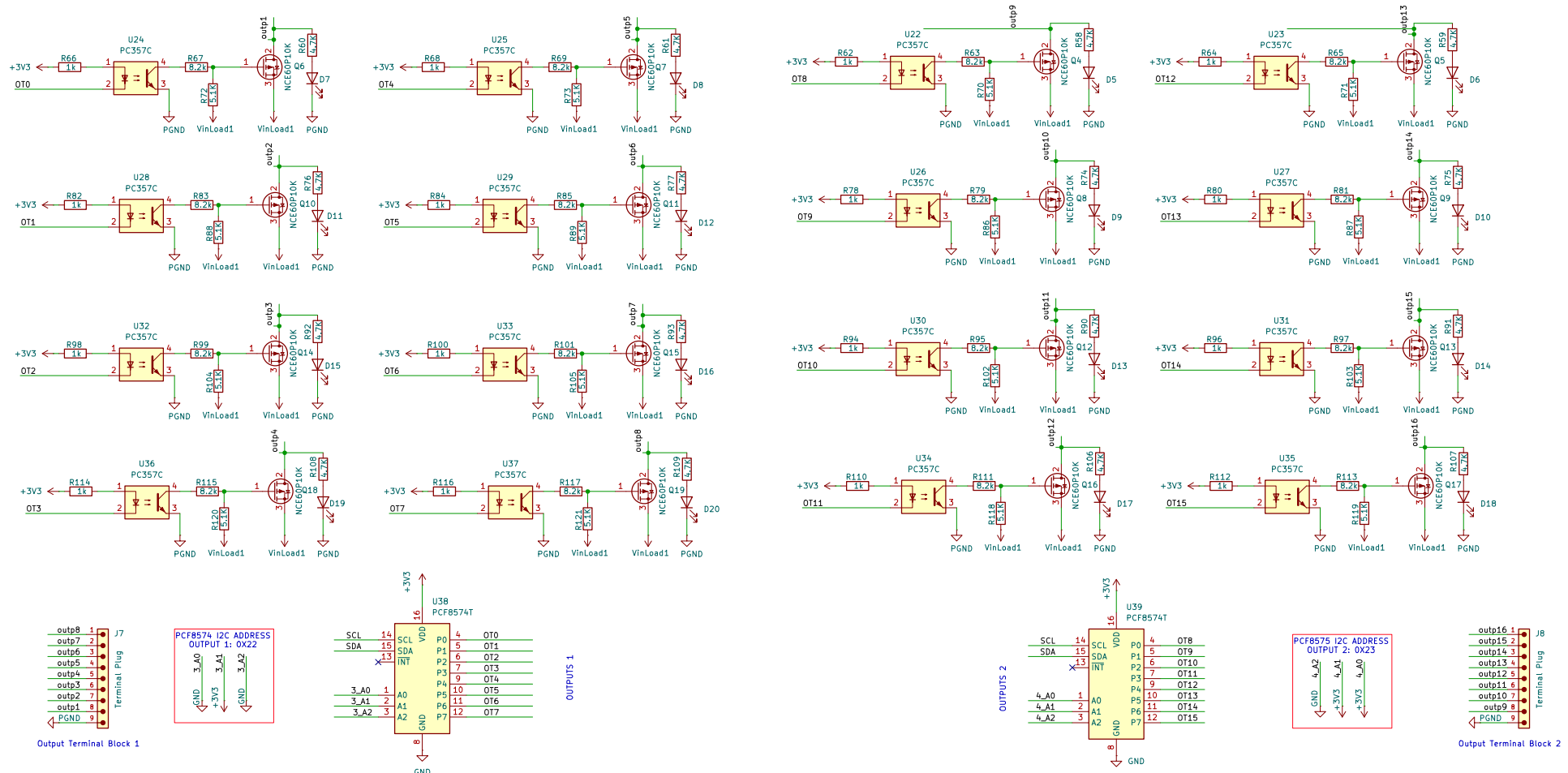
Tivana Holdings
Sheet: /16 DIGITAL INPUTS/
File: 16_digital_inputs.kicad_sch

Title: TSC BW 16 V3.1

Size: A3 Date: KICad E.D.A. 10.0.5

Rev: V3.1
Id: 4/17

16 ISOLATED OUTPUTS



PCF Pull Up Resistors

Designed by: Lucas Tivana

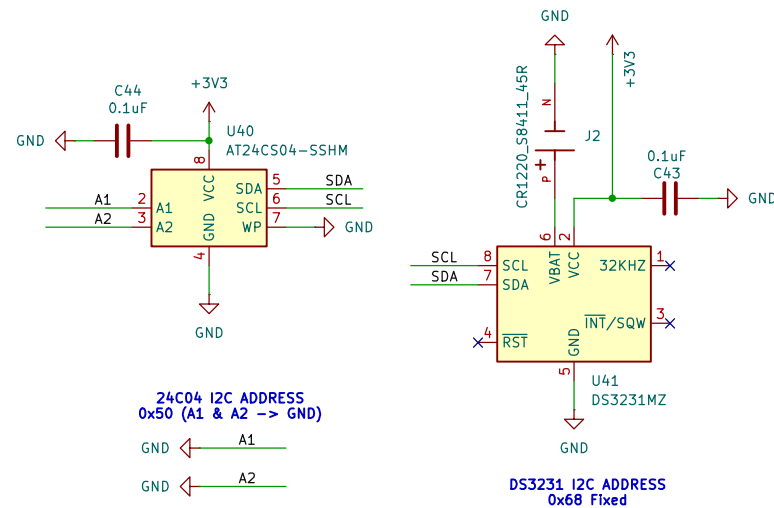
Tivana Holdings
Sheet: /16 DIGITAL OUTPUTS/
File: 16_digitalOutputs.kicad_sch

Title: TSC BW 16 V3.1

Size: A3 Date: Kicad E.D.A. 10.0.5

Rev: V3.1
Id: 5/17

RTC DS3231 & 24C04 EEPROM



GLOBAL PINS / LABELS

SCL

SDA

Designed by: Lucas Tivana

Tivana Holdings

Sheet: /RTC/

File: rtc.kicad_sch

Title: TSC BW 16 V3.1

Size: A4

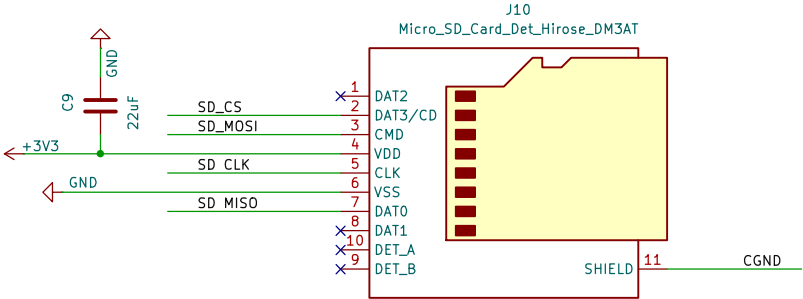
Date:

KiCad E.D.A. 10.0.5

Rev: V3.1

Id: 6/17

SD CARD



GLOBAL PINS / LABELS

SD_CS

SD_CS

SD_MOSI

SD_MOSI

SD_CLK

SD_CLK

SD_MISO

SD_MISO

SD Card (SPI)

MOSI – PIN20 : GPIO12

SCK – PIN21 : GPIO13

MISO – PIN22 : GPIO14

CS – PIN19 : GPIO11

SD-CD – PIN23 : GPIO21

Designed by: Lucas Tivana

Tivana Holdings

Sheet: /SD CARD/
File: sd_card.kicad_sch

Title: TSC BW 16 V3.1

Size: A4

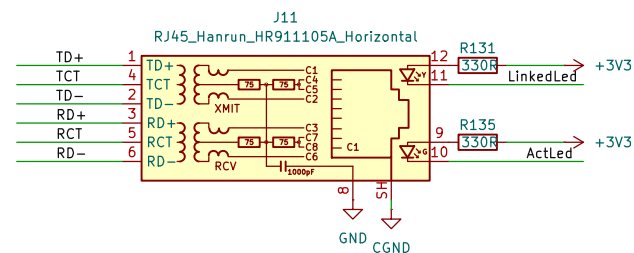
Date:

Rev: V3.1

KiCad E.D.A. 10.0.5

Id: 7/17

EMI FILTER



SCS	ETH_SCS
SCLK	ETH_SCLK
MISO	ETH_MISO
MOSI	ETH_MOSI
INT	ETH_INT
RST	ETH_RST

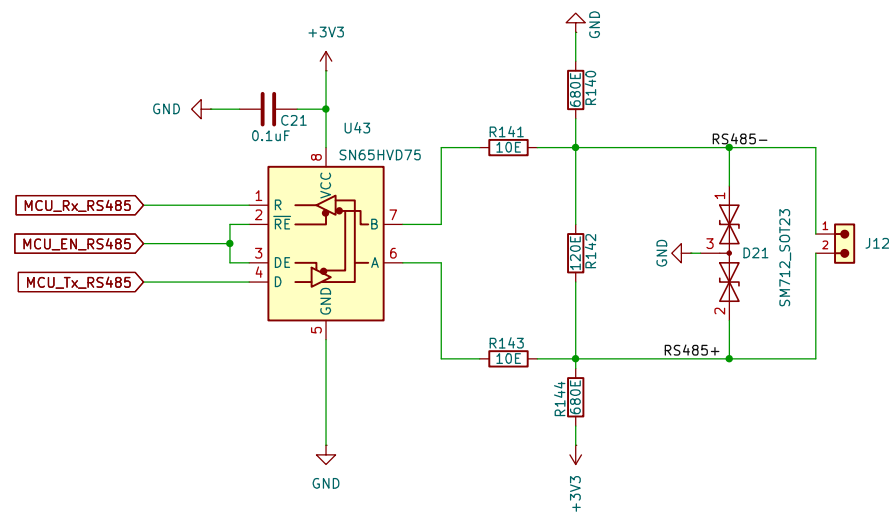
```

ethernet:
  type: W5500
  clk_pin: GPIO42
  mosi_pin: GPIO43
  miso_pin: GPIO44
  cs_pin: GPIO15
  interrupt_pin: GPIO2
  reset_pin: GPIO1

```

Id: 8/17

RS 485 V1



```
#define RS485_EN 21

void rs485Transmit()
digitalWrite(RS485_EN, HIGH);

void rs485Receive()
digitalWrite(RS485_EN, LOW);

void sendRS485(String msg)
{
  rs485Transmit();
  Serial2.print(msg);
  Serial2.flush();
  rs485Receive();
}
```

GLOBAL PINS / LABELS

485TX

485TX

485_EN

485_EN

485RX

485RX

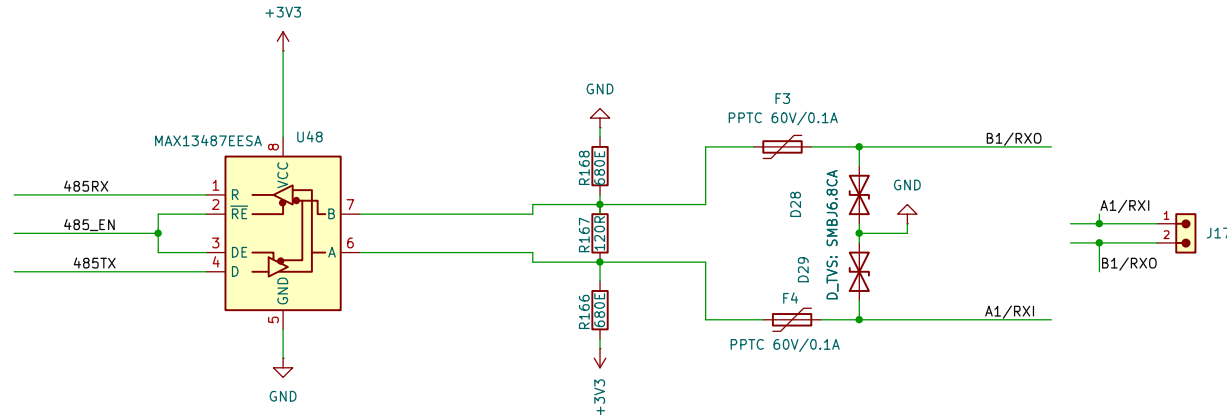
RS485

TXD - PIN9 : GPIO16

RXD - PIN10 : GPIO17

TXD - PIN23 : GPIO21

RS 485 V2



```
#define RS485_EN 21

void rs485Transmit()
{
    digitalWrite(RS485_EN, HIGH);

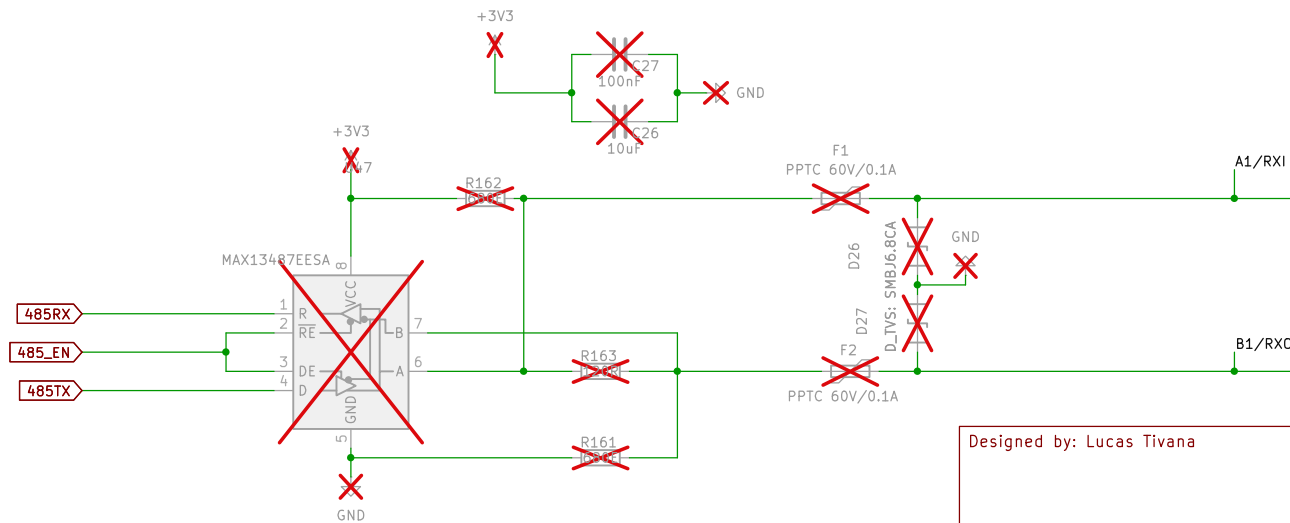
    void rs485Receive()
    {
        digitalWrite(RS485_EN, LOW);
    }

    void sendRS485(String msg)
    {
        rs485Transmit();
        Serial2.print(msg);
        Serial2.flush();
        rs485Receive();
    }
}
```

GLOBAL PINS / LABELS

485TX → 485TX
485_EN → 485_EN
485RX → 485RX

RS485
TXD - PIN9 : GPIO16
RXD - PIN10 : GPIO17
TXD - PIN23 : GPIO21



Designed by: Lucas Tivana

Tivana Holdings

Sheet: /RS485 v2/

File: RS485 v2.kicad_sch

Title: TSC BW 16 V3.1

Size: A4

Date:

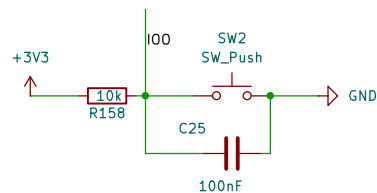
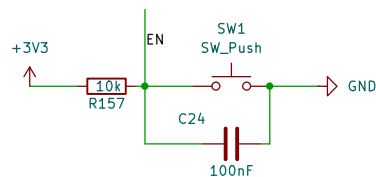
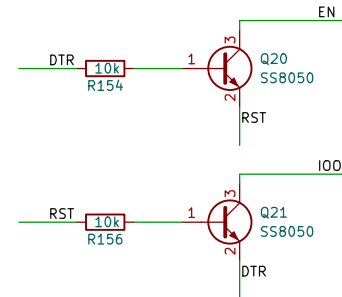
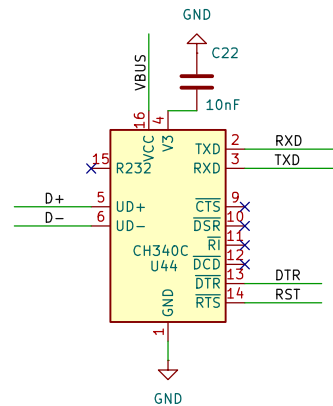
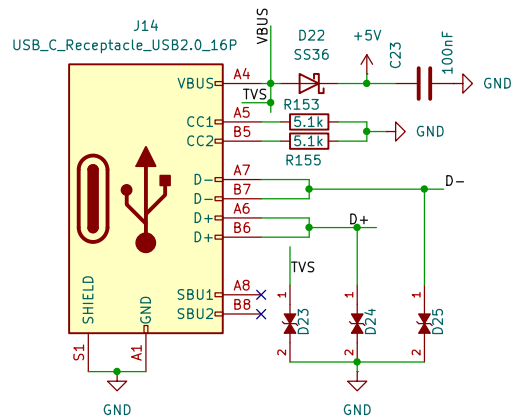
KiCad E.D.A. 10.0.5

Rev: V3.1

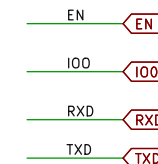
Id: 10/17

USB – TO – UART

AUTO PROGRAM



GLOBAL PINS / LABELS



USB C – UART
 EN – PIN3 : NO GPIO
 IOO – PIN27 : GPIO0
 RXD – PIN36/RXD0 : GPIO44,U0RXD
 TXD – PIN37/ TXD0 : GPIO43,U0TXD

Designed by: Lucas Tivana
 USB Type C to UART
 Auto Program

Tivana Holdings

Sheet: /USB_C to UART/
 File: usb_c_to_uart.kicad_sch

Title: TSC BW 16 V3.1

Size: A4

Date:

KiCad E.D.A. 10.0.5

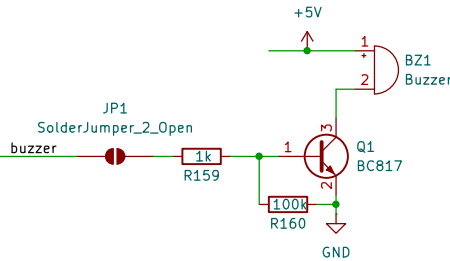
Rev: V3.1

Id: 11/17

1	2	3	4	5	6
A					A
B					B
C					C
D					D
1	2	3	4	5	6

Designed by: Lucas Tivana		
Tivana Holdings		
Sheet: /External Connections/ File: external_connections.kicad_sch		
Title: TSC BW 16 V3.1		
Size: A4	Date:	Rev: V3.1
KiCad E.D.A. 10.0.5		Id: 12/17

BUZZER



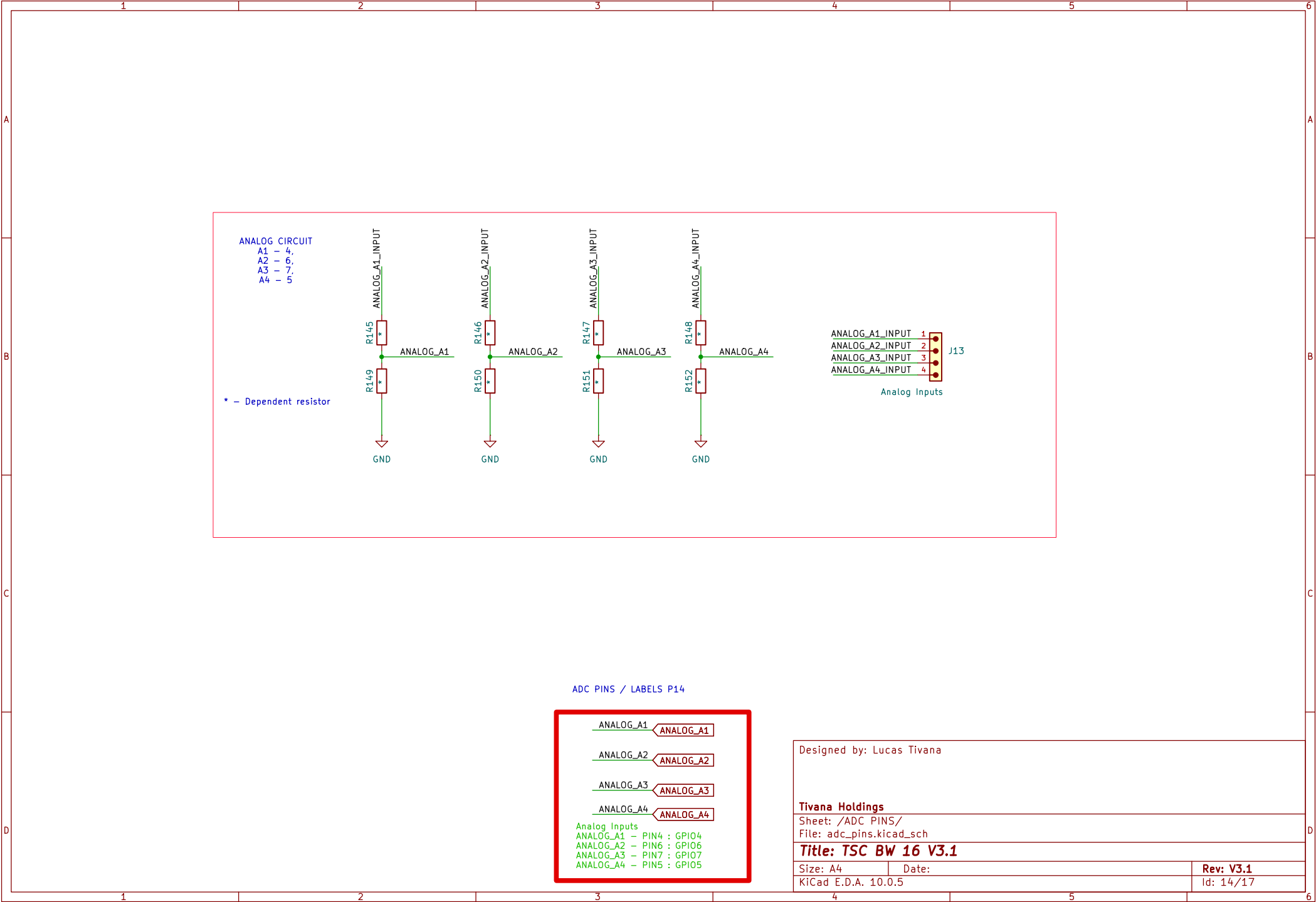
GLOBAL PINS / LABELS

buzzer

BUZZER

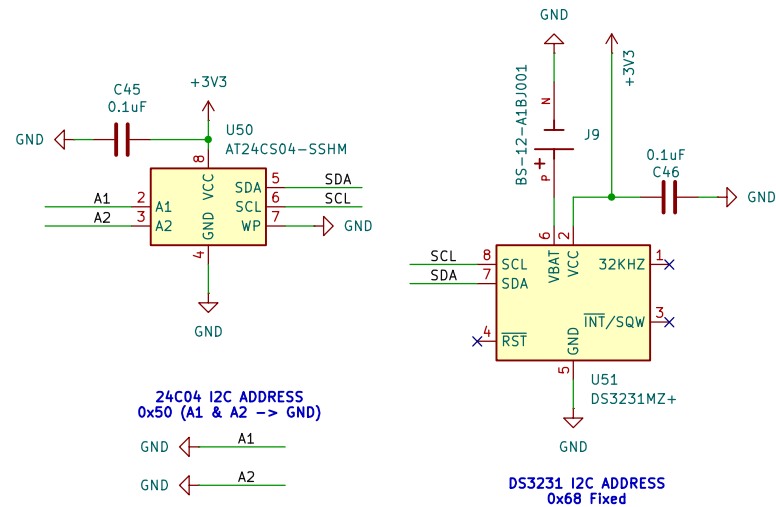
BUZZER – PIN25 : GPIO48

Sheet: /Buzzer/		
File: Buzzer.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.5	Id: 13/17	



RTC 2

DS3231MZ+ & 24C04 EEPROM



GLOBAL PINS / LABELS

SCL

SDA

Sheet: /RTC2/
File: RTC2.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.5

Rev:

Id: 16/17

